

[Vitamin C: Fact Sheet for Consumers \(original version\)](#)

Vitamin C: Fact Sheet for Consumers

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For information on vitamin C and COVID-19, see [Dietary Supplements in the Time of COVID-19](#).

What is vitamin C and what does it do?

Vitamin C, also known as ascorbic acid, is a water-soluble nutrient found in some foods. In the body, it acts as an antioxidant, helping to protect cells from the damage caused by free radicals. Free radicals are compounds formed when our bodies convert the food we eat into energy. People are also exposed to free radicals in the environment from cigarette smoke, air pollution, and ultraviolet light from the sun.

The body also needs vitamin C to make collagen, a protein required to help wounds heal. In addition, vitamin C improves the absorption of iron from plant-based foods and helps the immune system work properly to protect the body from disease.

How much vitamin C do I need?

The amount of vitamin C you need each day depends on your age. Average daily recommended amounts for different ages are listed below in milligrams (mg).

Life Stage	Recommended Amount
Birth to 6 months	40 mg
Infants 7–12 months	50 mg
Children 1–3 years	15 mg
Children 4–8 years	25 mg
Children 9–13 years	45 mg
Teens 14–18 years (boys)	75 mg
Teens 14–18 years (girls)	65 mg
Adults (men)	90 mg
Adults (women)	75 mg
Pregnant teens	80 mg
Pregnant women	85 mg

Life Stage	Recommended Amount
Breastfeeding teens	115 mg
Breastfeeding women	120 mg

If you smoke, add 35 mg to the above values to calculate your total daily recommended amount.

What foods provide vitamin C?

Fruits and vegetables are the best sources of vitamin C. You can get recommended amounts of vitamin C by eating a variety of foods including the following:

- Citrus fruits (such as oranges and grapefruit) and their juices as well as red and green pepper and kiwifruit, which have a lot of vitamin C.
- Other fruits and vegetables—such as broccoli, strawberries, cantaloupe, baked potatoes, and tomatoes—which also have vitamin C.
- Some foods and beverages that are fortified with vitamin C. To find out if vitamin C has been added to a food product, check the product labels.

The vitamin C content of food may be reduced by prolonged storage and by cooking. Steaming or microwaving may lessen cooking losses. Fortunately, many of the best food sources of vitamin C, such as fruits and vegetables, are usually eaten raw.

What kinds of vitamin C dietary supplements are available?

Most multivitamins have vitamin C. Vitamin C is also available alone as a dietary supplement or in combination with other nutrients. The vitamin C in dietary supplements is usually in the form of ascorbic acid, but some supplements have other forms, such as sodium ascorbate, calcium ascorbate, other mineral ascorbates, and ascorbic acid with bioflavonoids. Research has not shown that any form of vitamin C is better than the other forms.

Am I getting enough vitamin C?

Most people in the United States get enough vitamin C from foods and beverages. However, certain groups of people are more likely than others to have trouble getting enough vitamin C:

- People who smoke and those who are exposed to secondhand smoke, in part because smoke increases the amount of vitamin C that the body needs to repair damage caused by free radicals. People who smoke need 35 mg more vitamin C per day than nonsmokers.
- Infants who are fed evaporated or boiled cow's milk because cow's milk has very little vitamin C and heat can destroy vitamin C. Cow's milk is not recommended for infants under 1 year of age. Breast milk and infant formula have adequate amounts of vitamin C.
- People who eat a very limited variety of food.
- People with certain medical conditions such as severe malabsorption, some types of cancer, and kidney disease requiring hemodialysis.

What happens if I don't get enough vitamin C?

Vitamin C deficiency is rare in the United States and Canada. People who get little or no vitamin C (below about 10 mg per day) for many weeks can get scurvy. Scurvy causes fatigue, inflammation of the gums, small red or purple spots on the skin, joint pain, poor wound healing, and corkscrew hairs. Additional signs of scurvy include depression as well as swollen, bleeding gums and loosening or loss of teeth. People with scurvy can also develop anemia. Scurvy is fatal if it is not treated.

What are some effects of vitamin C on health?

Scientists are studying vitamin C to understand how it affects health. Here are several examples of what this research has shown.

Cancer prevention and treatment

People with high intakes of vitamin C from fruits and vegetables might have a lower risk of getting many types of cancer, such as lung, breast, and colon cancer. However, taking vitamin C supplements, with or without other antioxidants, doesn't seem to protect people from getting cancer.

It is not clear whether taking high doses of vitamin C is helpful as a treatment for cancer. Vitamin C's effects appear to depend on how it is administered to the patient. Oral doses of vitamin C can't raise blood levels of vitamin C nearly as high as intravenous doses given through injections. A few studies in animals and test tubes indicate that very high blood levels of vitamin C might shrink tumors. However, more research is needed to determine whether high-dose intravenous vitamin C helps treat cancer in people.

Vitamin C dietary supplements and other antioxidants might interact with chemotherapy and radiation therapy for cancer. People being treated for cancer should talk with their oncologist before taking vitamin C or other antioxidant supplements, especially in high doses.

Cardiovascular disease

People who eat lots of fruits and vegetables seem to have a lower risk of cardiovascular disease. Researchers believe that the antioxidant content of these foods might be partly responsible for this association because oxidative damage is a major cause of cardiovascular disease. However, scientists aren't sure whether vitamin C itself, either from food or supplements, helps protect people from cardiovascular disease. It is also not clear whether vitamin C helps prevent cardiovascular disease from getting worse in people who already have it.

Age-related macular degeneration and cataracts

Age-related macular degeneration (AMD) and cataracts are two of the leading causes of vision loss in older people. Researchers do not believe that vitamin C and other antioxidants affect the risk of getting AMD. However, research suggests that vitamin C combined with other nutrients might help slow AMD progression.

In a large study among older people with AMD who were at high risk of developing advanced AMD, those who took a daily dietary supplement with 500 mg vitamin C, 80 mg zinc, 400 International Units (IU) vitamin E, 15 mg beta-carotene, and 2 mg copper for about 6 years had a lower chance of developing advanced AMD. They also had less vision loss than those who did not take the dietary supplement. People who have or are developing the disease might want to talk with their doctor about taking dietary supplements.

The relationship between vitamin C and cataract formation is unclear. Some studies show that people who get more vitamin C from foods have a lower risk of getting cataracts. However, further research is needed to clarify this association and to determine whether vitamin C supplements affect the risk of getting cataracts.

The common cold

Although vitamin C has long been a popular remedy for the common cold, research shows that for most people, vitamin C supplements do not reduce the risk of getting the common cold. However, people who take vitamin C supplements regularly might have slightly shorter colds or somewhat milder symptoms when they do have a cold. Using vitamin C supplements after cold symptoms start does not appear to be helpful.

Can vitamin C be harmful?

Taking too much vitamin C can cause diarrhea, nausea, and stomach cramps. In people with a condition called hemochromatosis, which causes the body to store too much iron, high doses of vitamin C could

worsen iron overload and damage body tissues.

The daily upper limits for vitamin C include intakes from all sources—food, beverages, and supplements—and are listed below.

Life Stage	Upper Limit
Birth to 12 months	Not established
Children 1–3 years	400 mg
Children 4–8 years	650 mg
Children 9–13 years	1,200 mg
Teens 14–18 years	1,800 mg
Adults	2,000 mg

Does vitamin C interact with medications or other dietary supplements?

Vitamin C dietary supplements can interact or interfere with medicines that you take. Here are several examples:

- Vitamin C dietary supplements might interact with cancer treatments, such as chemotherapy and radiation therapy. It is not clear whether vitamin C might have the unwanted effect of protecting tumor cells from cancer treatments or whether it might help protect normal tissues from getting damaged. If you are being treated for cancer, check with your health care provider before taking vitamin C or other antioxidant supplements, especially in high doses.
- In one study, vitamin C plus other antioxidants (such as vitamin E, selenium, and beta-carotene) reduced the heart-protective effects of two drugs taken in combination (a statin and niacin) to control blood-cholesterol levels. It is not known whether this interaction also occurs with other statins. Health care providers should monitor lipid levels in people taking both statins and antioxidant supplements.

Tell your doctor, pharmacist, and other health care providers about any dietary supplements and medicines you take. They can tell you if those dietary supplements might interact or interfere with your prescription or over-the-counter medicines or if the medicines might interfere with how your body absorbs, uses, or breaks down nutrients.

Vitamin C and healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's *Dietary Guidelines for Americans*. Foods contain vitamins, minerals, dietary fiber, and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible to meet needs for one or more nutrients (for example, during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the [Dietary Guidelines for Americans](#).

Where can I find out more about vitamin C?

- For general information on vitamin C
 - Office of Dietary Supplements (ODS) Health Professional Fact Sheet on [Vitamin C](#)
 - [Vitamin C](#), MedlinePlus
- For more information on food sources of vitamin C
 - USDA's [FoodData Central](#)
 - Nutrient List for vitamin C (listed by [food](#) or by [vitamin C content](#)), USDA
- For more advice on choosing dietary supplements
 - ODS [Frequently Asked Questions: Which brand\(s\) of dietary supplements should I purchase?](#)

- For information about building a healthy dietary pattern
 - [Dietary Guidelines for Americans](#)

Glossary

absorption

In nutrition, the process of moving protein, carbohydrates, fats, and other nutrients from the digestive system into the bloodstream. Most absorption occurs in the small intestine.

age-related macular degeneration

AMD. An eye disease that results in a loss of central, "straight-ahead" vision. AMD is the leading cause of vision loss in older Americans.

anemia

A condition in which the number of red blood cells in the blood, or the amount of hemoglobin in them, is lower than normal, causing a condition in which red blood cells are not able to supply enough oxygen to all the tissues in the body. Hemoglobin is the substance in red blood cells that carries oxygen to your body's cells.

antioxidant

A substance that protects cells from damage caused by free radicals (compounds formed during the metabolism of oxygen). It may help prevent the development of some chronic diseases such as cancer. Antioxidants include beta-carotene; lutein; lycopene; vitamins A, C, and E; selenium; and zinc.

association

A relationship between two conditions or states such that if one is present, the other is likely to be present as well. An association between two conditions or states, however, does not necessarily imply a cause and effect relationship. The terms association and relationship are often used interchangeably.

beta-carotene

A carotenoid found in carrots, cantaloupe, apricots, sweet potatoes, pumpkin, winter squash, mangos, collard greens, spinach, kale, broccoli, and other orange, red, and dark green fruits and vegetables.

calcium

A mineral found throughout the body. Calcium is needed for healthy bones and teeth, for nerves and enzymes to function properly, and for blood clotting. Calcium is found in some foods, including milk, yogurt, and cheese, and in Chinese cabbage, kale, broccoli and fortified foods, such as many drinks, tofu, and cereals.

cancer

A group of diseases in which cells divide abnormally and without control, and spread to nearby tissues and other parts of the body. Without treatment, cancer can stop organs from working normally, damage body systems, and cause the patient to die. Cancer may be caused by multiple factors, such as radiation, sunlight, tobacco, certain viruses, and poisonous chemicals; however, the cause of many cancers is unknown.

cardiovascular disease

CVD. A general term referring to disorders of the heart and blood vessels. CVD includes coronary artery disease, heart failure, atherosclerosis, high blood pressure, peripheral artery disease, and stroke.

cataract

A condition in which the lens of the eye becomes cloudy. Symptoms include blurred, cloudy, or double vision; sensitivity to light; and difficulty seeing at night. Without treatment, cataracts can cause blindness. Cataracts occur in people of all ages but are most common in the elderly.

cell

The individual unit that makes up the tissues of the body. All living things are made up of one or more cells, which are the smallest units of living structure capable of independent existence.

chemotherapy

A chemical that kills bacteria, viruses, fungi, or tumor cells. It usually refers to drugs used in cancer treatment.

cholesterol

A substance found throughout the body. It is made by the liver and is an important component of cells. Cholesterol is also used to make hormones, bile acid, and vitamin D. Foods that come from animals contain cholesterol, including eggs, dairy products, meat, poultry and fish. High blood levels of cholesterol increase a person's chance (risk) of developing atherosclerosis and heart disease.

collagen

A strong, flexible protein found in cartilage, tendons, bone, skin, and other connective tissue.

colon cancer

Uncontrolled growth of abnormal cells in the large intestine (the tube-like organ connected to the small intestine at one end and the anus at the other).

common cold

A nose and throat infection caused by a virus. Symptoms include runny nose, sneezing, congestion, sore throat, and cough.

compound

In pharmacy, a substance that contains more than one ingredient.

control

In a clinical trial, the group of participants that does not receive the new treatment being studied. This group is compared with the group receiving the new treatment, to see whether the new treatment works. In an observational study, the controls are participants who do not have a particular health condition; the control group is compared with the group of participants who do have the condition to see if certain factors (such as diet, activity level, or use of dietary supplements) may be associated with developing or preventing the condition.

copper

In nutrition, a mineral the body needs (along with iron) to make red blood cells. Copper also helps keep the immune system, blood vessels, nerves, and bones healthy. Copper is found in some foods, including oysters and other shellfish, whole grains, beans, nuts, potatoes, organ meats, dark leafy greens, and dried fruits.

deficiency

An amount that is not enough; a shortage.

depression

A serious medical condition that can interfere with an individual's ability to work, study, sleep, and eat. Symptoms include ongoing feelings of sadness and despair, loss of energy, and loss of interest or pleasure in activities that were once enjoyed. A person who has depression may experience changes in eating or sleeping habits, and have thoughts of death or suicide.

diarrhea

Loose, watery stools.

dietary fiber

A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

Dietary Guidelines for Americans

Advice from the federal government to promote health and reduce the chance (risk) of long-lasting (chronic) diseases through nutrition and physical activity. The Guidelines are updated and published every 5 years by the US Department of Health and Human Services and the US Department of Agriculture.

dietary supplement

A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

dose

The amount of medicine or other substance taken at one time or over a specific period of time.

drug

Any substance (other than food) that is used to prevent, diagnose, treat, or relieve symptoms of a disease or abnormal condition. Also, a substance that alters mood or body function or that can be habit-forming or addictive, especially a narcotic.

fatal

Deadly; causing death.

fatigue

Extreme tiredness and an inability to function due to lack of energy.

fortified

When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many

breakfast cereals are "fortified" with several vitamins and minerals.

free radical

An atom or molecule made in the body that can damage cells. A free radical has at least one unpaired electron, which makes it unstable. To become stable, the free radical takes an electron away from another atom, which makes *that* atom unstable, and starts a chain reaction that can injure cells. Free radicals are made during chemical changes that take place in a cell or an organism to produce energy and basic materials needed for important life processes (metabolism). They also come from tobacco smoke, pollution, radiation from the sun and x-rays, and other sources outside the body. Free radicals damage cells, cause genetic alterations (mutations), and may play a role in cancer, heart disease, and age-related diseases (such as Alzheimer's, Parkinson's, and Lou Gehrig's diseases). Free radicals are also beneficial; they are involved in killing germs (microorganisms) and they help hormones and chemical messengers communicate with cells. Proteins (enzymes) made by the body, and vitamin C, vitamin E, and beta carotene in the diet help prevent free radical damage.

health care provider

A person who supplies health care services. Health care providers include individuals with professional training (including doctors, nurses, technicians, and aides).

hemochromatosis

A condition in which the body absorbs more iron than it needs and stores it in the liver, heart, and pancreas. Hemochromatosis causes liver disease, heart problems, and organ failure.

hemodialysis

The use of a machine to remove wastes and extra fluid from the blood when the kidneys have stopped working. The cleaned blood is then returned to the body.

immune system

A group of organs and cells that defends the body against infection, disease, and altered (mutated) cells. It includes the thymus, spleen, lymphatic system (lymph nodes and lymph vessels), bone marrow, tonsils, and white blood cells.

infant

A child younger than 12 months old.

infant formula

An artificial form of breast milk.

inflammation

Redness, swelling, pain, and/or a feeling of heat in an area of the body. It is a protective reaction to injury, disease, or irritation of tissues.

injection

Use of a syringe and needle to push fluids or drugs into the body. Also called a shot.

interaction

A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

intravenous

Into or within a vein, such as an intravenous injection.

iron

In nutrition, a mineral the body needs to make red blood cells, proteins, and enzymes; and for the control of cell growth and cell specialization. Iron is found in some foods, including red meats, fish, poultry, lentils, and beans.

IU

International Unit. A measurement used to measure the activity of some vitamins and other biological substances (such as enzymes and hormones).

kidney disease

A condition that lessens the ability of the kidneys to filter wastes from the blood, keep blood chemical levels balanced, and make certain hormones. The two most common causes of kidney disease are diabetes and high blood pressure. People with long-term kidney disease may need dialysis or a kidney transplant to stay alive.

label

When referring to dietary supplements, information that appears on the product container, including a descriptive name of the product stating that it is a "supplement"; the name and place of business of the manufacturer, packer, or distributor; a complete list of ingredients; and each dietary ingredient contained in the product. Supplements must also include directions for use, nutrition labeling in the

form of a Supplement Facts panel that identifies each dietary ingredient contained in the product and the serving size, amount, and active ingredients.

lot

A batch, or a specific identified portion of a batch, having uniform character and quality within specified limits; or, an amount produced in a unit of time or quantity.

lung

An organ in the chest that supplies oxygen to the body and removes carbon dioxide. There are two lungs in the body.

malabsorption

A reduced ability to properly absorb nutrients. It can be caused by injury to the digestive tract, a genetic disease, or other conditions. Malabsorption can lead to malnutrition.

milligram

mg. A measure of weight. It is a metric unit of mass equal to 0.001 gram (it weighs 28,000 times less than an ounce).

mineral

In nutrition, an inorganic substance found in the earth that is required to maintain health.

multivitamin/mineral dietary supplement

MVM. A product that is meant to supplement the diet. MVMs contain a variety of vitamins and minerals. The number and amounts of these nutrients can vary substantially by product.

nausea

The uneasy feeling of having an urge to throw up (vomit).

niacin

A nutrient that is needed by the body to make energy from food, and for important cellular functions. Niacin is found in some foods, including poultry, beef, pork, fish, nuts, legumes, whole grains, and enriched or fortified breads and cereals. In medicine it is used as a drug to lower high blood cholesterol levels. It is one of the B vitamins. Also called nicotinamide and nicotinic acid.

nutrient

A chemical compound in food that is used by the body to function and maintain health. Examples of nutrients include proteins, fats, carbohydrates, vitamins, and minerals.

Office of Dietary Supplements

ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

oral

By mouth; having to do with the mouth.

oxidative damage

Changes that take place in the body's cells as a normal result of living (such as from eating food or being exposed to sunlight). Too many of these chemical changes may increase the risk of chronic diseases like cancer, heart disease, diabetes, and age-related eye disease. Antioxidants help to protect the body from excess oxidative damage.

pharmacist

A person licensed to make and dispense (give out) prescription drugs and who has been taught how they work, how to use them, and their side effects.

prescription

A written order from a health care provider for medicine, therapy, or tests.

prevent

To stop from happening.

progression

In medicine, the course of a disease as it becomes worse. For example, as cancer progresses, it spreads in the body.

protein

A molecule made up of amino acids that the body needs for good health. Proteins are the basis of body structures such as skin and muscle, and substances such as enzymes and antibodies.

radiation therapy

The use of high-energy radiation from x-rays and other sources to kill cancer cells, shrink tumors, and treat other conditions.

risk

The chance or probability that a harmful event will occur. In health, for example, the chance that someone will develop a disease or condition.

scurvy

A disease caused by a lack of vitamin C in the diet. Symptoms include anemia, gum disease, bleeding, and bruising.

secondhand smoke

Smoke that comes from the burning end of a cigarette, cigar, or pipe and smoke exhaled by a smoker. Inhaling secondhand smoke can cause cancer, respiratory tract infections, and heart disease.

selenium

A mineral required in very small amounts to make important enzymes that are essential for good health. Selenium is found in some foods, including plant foods grown in selenium-rich soil, and some meats and seafood.

sign

An indication of disease that can be seen and/or measured. Examples include high fever, high blood pressure, infection, and coughing up blood.

statin

A drug used to treat high cholesterol. Statins lower the amount of cholesterol and certain fats in the blood.

supplement

A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

symptom

A feeling of sickness that an individual can sense, but that cannot be measured by a healthcare professional. Examples include headache, tiredness, stomach ache, depression, and pain.

tissue

A group or layer of cells in a living organism that work together to perform a specific function.

treat

To care for a patient with a disease by using medicine, surgery, or other approaches.

tumor

An abnormal growth of tissue that forms when cells divide more often than they should. Tumors may be benign (not cancer) or malignant (cancer).

ultraviolet light

Invisible rays that are part of the energy that comes from the sun. Ultraviolet light also comes from sun lamps and tanning beds. It can damage the skin and cause skin cancer.

upper limit

UL. The largest daily intake of a nutrient considered safe for most people. Taking more than the UL is not recommended and may be harmful. The UL for each nutrient is set by the Food and Nutrition Board at the National Academies of Sciences, Engineering, and Medicine. For example, the UL for vitamin A is 3,000 micrograms/day. Women who consume more than this amount every day shortly before or during pregnancy have an increased chance (risk) of having a baby with a birth defect. Also called the tolerable upper intake level.

US Department of Agriculture

USDA promotes America's health through food and nutrition, and advances the science of nutrition by monitoring food and nutrient consumption and updating nutrient requirements and food composition data. USDA is responsible for food safety, improving nutrition and health by providing food assistance and nutrition education, expanding markets for agricultural products, managing and protecting US public and private lands, and providing financial programs to improve the economy and quality of rural American life.

vitamin

A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

vitamin C

A nutrient needed by the body to make collagen (a protein found in cartilage, tendons, ligaments, bone, and blood vessels), to absorb iron from food, and for wound healing. It is an antioxidant and protects cells from free radical damage. Vitamin C is found in some foods including citrus fruits, strawberries, peppers, dark green vegetables, tomatoes, and potatoes. Also called ascorbic acid.

vitamin E

A nutrient needed by the body to help keep the immune system healthy and to repair damage to DNA. It is an antioxidant that protects cells from free radical damage. Vitamin E is found in some foods, including vegetable oils, nuts and seeds, fortified breakfast cereals, and spinach, broccoli, kiwi, and mangos.

zinc

A mineral found in most cells of the body. It helps enzymes work properly, helps maintain a healthy immune system, helps maintain the senses of taste and smell, and is needed for wound healing, making DNA, and normal growth and development during pregnancy, childhood, and adolescence. Zinc is found in some foods, including oysters, red meat, poultry, beans, nuts, certain seafood, whole grains, fortified breakfast cereals, and dairy products.

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